

5. Uniform Distribution — Solved Problems in Data Science

1. What is Uniform Distribution?

A **Uniform distribution** describes a situation where every value in a specified interval has the **same probability density**.

For a continuous Uniform distribution:

$$X \sim U(a, b)$$

where:

- a = lower limit
- b = upper limit

The probability density function (PDF) is:

$$f(x) = \begin{cases} \frac{1}{b-a}, & a \leq x \leq b \\ 0, & \text{otherwise} \end{cases}$$

Mean

$$E(X) = \frac{a+b}{2}$$

Variance

$$Var(X) = \frac{(b-a)^2}{12}$$

Standard Deviation

$$\sigma = \frac{b-a}{\sqrt{12}}$$

Problem 1: Random Website Response Time

Suppose the response time of a web API is assumed to be uniformly distributed between **100 ms and 500 ms**.

Find the probability that a randomly selected request has a response time between **200 ms and 300 ms**.

Solution

Given:

$$X \sim U(100, 500)$$

We need:

$$P(200 \leq X \leq 300)$$

For a continuous Uniform distribution:

$$P(c \leq X \leq d) = \frac{d - c}{b - a}$$

Therefore:

$$\begin{aligned} P(200 \leq X \leq 300) &= \frac{300 - 200}{500 - 100} \\ P(200 \leq X \leq 300) &= \frac{100}{400} \end{aligned}$$

$$P(200 \leq X \leq 300) = 0.25$$

Answer

25%

Data Science Interpretation

There is a **25% probability** that a randomly selected API request has a response time between **200 ms and 300 ms**.

Problem 2: Customer Session Duration

Suppose customer session duration on a website is modeled uniformly between **5 minutes and 15 minutes**.

Find the probability that a randomly selected customer stays for **more than 10 minutes**.

Solution

Given:

$$X \sim U(5,15)$$

We need:

$$P(X > 10)$$

Therefore:

$$P(X > 10) = \frac{15 - 10}{15 - 5}$$
$$P(X > 10) = \frac{5}{10}$$

$$P(X > 10) = 0.5$$

Answer

50%

Interpretation

There is a **50% probability** that a customer stays on the website for more than 10 minutes.

Problem 3: Random Train-Test Split

A Data Scientist randomly selects a number between **0 and 1** to determine whether a record belongs to the training or testing process.

Let:

$$X \sim U(0,1)$$

Find the probability that the generated random number is between **0.2 and 0.7**.

Solution

We need:

$$P(0.2 \leq X \leq 0.7)$$

Using:

$$P(c \leq X \leq d) = \frac{d - c}{b - a}$$

Here:

$$a = 0, \quad b = 1$$

Therefore:

$$P(0.2 \leq X \leq 0.7) = \frac{0.7 - 0.2}{1 - 0}$$

$$P(0.2 \leq X \leq 0.7) = 0.5$$

Answer

50%

Data Science Application

Uniform random numbers between 0 and 1 are fundamental to:

- Random sampling
- Monte Carlo simulation
- Random initialization
- Train-test sampling
- Synthetic data generation

Problem 4: Machine Sensor Reading

A sensor produces measurements between **20°C and 30°C**, and all values within this range are assumed equally likely.

Find the probability that a measurement is between **24°C and 27°C**.

Solution

Given:

$$X \sim U(20,30)$$

We need:

$$P(24 \leq X \leq 27)$$

Therefore:

$$P(24 \leq X \leq 27) = \frac{27 - 24}{30 - 20}$$

$$P(24 \leq X \leq 27) = \frac{3}{10}$$

$$P(24 \leq X \leq 27) = 0.3$$

Answer

30%

Problem 5: Expected Website Response Time

A website's response time is uniformly distributed between **100 ms and 500 ms**.

Find the expected response time.

Solution

Given:

$$a = 100, \quad b = 500$$

For Uniform distribution:

$$E(X) = \frac{a + b}{2}$$

Therefore:

$$E(X) = \frac{100 + 500}{2}$$

$$E(X) = \frac{600}{2}$$

$$E(X) = 300 \text{ ms}$$

Answer

300 ms

Interpretation

The average response time under this Uniform model is **300 ms**.

Problem 6: Variance and Standard Deviation of Response Time

A server's response time is uniformly distributed between **100 ms** and **500 ms**.

Find:

1. Mean
2. Variance
3. Standard deviation

Solution

Given:

$$a = 100, \quad b = 500$$

Mean

$$E(X) = \frac{100 + 500}{2}$$

$$E(X) = 300$$

Variance

$$\begin{aligned} Var(X) &= \frac{(b - a)^2}{12} \\ Var(X) &= \frac{(500 - 100)^2}{12} \\ Var(X) &= \frac{400^2}{12} \\ Var(X) &= \frac{160000}{12} \end{aligned}$$

$$Var(X) = 13333.33$$

Standard Deviation

$$\sigma = \sqrt{13333.33}$$

$$\sigma \approx 115.47 \text{ ms}$$

Answer

Measure	Value
Mean	300 ms
Variance	13,333.33 ms ²
Standard deviation	115.47 ms

Problem 7: Probability of a Random Number

A machine-learning algorithm generates a random number:

$$X \sim U(0,10)$$

Find the probability that the generated number is **less than 3**.

Solution

We need:

$$P(X < 3)$$

For Uniform distribution:

$$P(X < 3) = \frac{3 - 0}{10 - 0}$$

$$P(X < 3) = \frac{3}{10}$$

$$P(X < 3) = 0.3$$

Answer

30%

Problem 8: Probability Above a Threshold

A feature value is uniformly distributed between **50 and 100**.

Find the probability that a randomly selected value is **greater than 80**.

Solution

Given:

$$X \sim U(50,100)$$

We need:

$$P(X > 80)$$

Therefore:

$$P(X > 80) = \frac{100 - 80}{100 - 50}$$

$$P(X > 80) = \frac{20}{50}$$

$$P(X > 80) = 0.4$$

Answer

40%

Data Science Interpretation

If the feature truly follows this Uniform model, **40% of observations are expected to exceed 80.**

Problem 9: Probability Within One Standard Deviation of the Mean

Suppose:

$$X \sim U(0,12)$$

Find the probability that X lies within one standard deviation of the mean.

Step 1: Mean

$$\mu = \frac{0 + 12}{2} = 6$$

Step 2: Standard deviation

$$\sigma = \frac{12 - 0}{\sqrt{12}}$$

$$\sigma = \frac{12}{\sqrt{12}}$$

$$\sigma \approx 3.464$$

One standard deviation from the mean is:

$$6 - 3.464 \leq X \leq 6 + 3.464$$

Therefore:

$$2.536 \leq X \leq 9.464$$

Step 3: Calculate probability

The interval length is:

$$9.464 - 2.536 = 6.928$$

The total interval length is:

12

Therefore:

$$P(|X - \mu| \leq \sigma) = \frac{6.928}{12}$$

$$P \approx 0.5774$$

Answer

57.74%

Interesting Point

For a Uniform distribution, approximately **57.74%** of observations fall within one standard deviation of the mean.

This differs from the familiar **68%** associated with a Normal distribution.

Problem 10: Random Initialization in Machine Learning

A neural network initializes a parameter X uniformly between:

-0.5 and 0.5

Find the probability that the initialized parameter is between **-0.2 and 0.3**.

Solution

Given:

$$X \sim U(-0.5, 0.5)$$

We need:

$$P(-0.2 \leq X \leq 0.3)$$

Therefore:

$$\begin{aligned} P(-0.2 \leq X \leq 0.3) &= \frac{0.3 - (-0.2)}{0.5 - (-0.5)} \\ P(-0.2 \leq X \leq 0.3) &= \frac{0.5}{1} \end{aligned}$$

$$P = 0.5$$

Answer

$$50\%$$

Data Science Application

Uniform distributions are commonly used in **simulation and random parameter initialization**, although modern neural-network initialization methods often use distributions specifically designed to maintain suitable activation and gradient scales.

Problem 11: A/B Testing — Random Assignment

Suppose a random number X is generated uniformly between 0 and 1.

A user is assigned to:

- Group A if $0 \leq X < 0.5$
- Group B if $0.5 \leq X \leq 1$

Find the probability that a user is assigned to Group A.

Solution

We need:

$$P(X < 0.5)$$

Since:

$$X \sim U(0,1)$$

we have:

$$P(X < 0.5) = \frac{0.5 - 0}{1 - 0}$$

$$P(X < 0.5) = 0.5$$

Answer

50%

Therefore, approximately half of the users are expected to be assigned to each group.

Problem 12: Data Sampling

A Data Scientist randomly generates an index between **1 and 101** using a continuous Uniform model.

What is the probability that the generated value falls between **20 and 50**?

Solution

Given:

$$X \sim U(1,101)$$

We need:

$$P(20 \leq X \leq 50)$$

Therefore:

$$P(20 \leq X \leq 50) = \frac{50 - 20}{101 - 1}$$
$$P(20 \leq X \leq 50) = \frac{30}{100}$$

$$P = 0.30$$

Answer

30%

Problem 13: Percentile Using Uniform Distribution

Suppose:

$$X \sim U(10,50)$$

Find the value corresponding to the **75th percentile**.

Solution

For a Uniform distribution, the cumulative distribution function is:

$$F(x) = \frac{x - a}{b - a}$$

At the 75th percentile:

$$F(x) = 0.75$$

Therefore:

$$\begin{aligned}\frac{x - 10}{50 - 10} &= 0.75 \\ \frac{x - 10}{40} &= 0.75 \\ x - 10 &= 30\end{aligned}$$

$$x = 40$$

Answer

75th percentile = 40

Interpretation

Approximately **75% of observations are below 40** under this Uniform model.

Problem 14: Finding the Probability Density

A variable representing a simulated feature is uniformly distributed between **2 and 8**.

Find its probability density function.

Solution

Given:

$$a = 2, \quad b = 8$$

For Uniform distribution:

$$f(x) = \frac{1}{b - a}$$

Therefore:

$$f(x) = \frac{1}{8 - 2}$$

$$f(x) = \frac{1}{6}$$

Thus:

$$f(x) = \begin{cases} \frac{1}{6}, & 2 \leq x \leq 8 \\ 0, & \text{otherwise} \end{cases}$$

Important Point

For a **continuous** distribution, the PDF value itself is not a probability.

For example:

$$f(5) = \frac{1}{6}$$

does **not** mean:

$$P(X = 5) = \frac{1}{6}$$

In fact, for a continuous random variable:

$$P(X = 5) = 0$$

Probabilities are obtained from **intervals/areas under the PDF**.

Problem 15: Real Data Science Problem — Simulation

A Data Scientist wants to simulate customer waiting times between **2 minutes and 8 minutes**, assuming every value in this interval is equally likely.

Let:

$$X \sim U(2,8)$$

Find:

1. Mean waiting time
 2. Variance
 3. Standard deviation
 4. Probability that waiting time is between 3 and 5 minutes
 5. Probability that waiting time exceeds 6 minutes
-

Solution

Given:

$$a = 2, \quad b = 8$$

1. Mean

$$E(X) = \frac{a+b}{2}$$
$$E(X) = \frac{2+8}{2}$$

$$E(X) = 5 \text{ minutes}$$

2. Variance

$$\text{Var}(X) = \frac{(b-a)^2}{12}$$

$$\begin{aligned} \text{Var}(X) &= \frac{(8-2)^2}{12} \\ \text{Var}(X) &= \frac{36}{12} \end{aligned}$$

$$\text{Var}(X) = 3$$

3. Standard Deviation

$$\sigma = \sqrt{3}$$

$$\sigma \approx 1.732 \text{ minutes}$$

4. Probability between 3 and 5 minutes

$$\begin{aligned} P(3 \leq X \leq 5) &= \frac{5-3}{8-2} \\ P(3 \leq X \leq 5) &= \frac{2}{6} \end{aligned}$$

$$P(3 \leq X \leq 5) = 0.3333$$

Therefore:

$$33.33\%$$

5. Probability of waiting more than 6 minutes

$$\begin{aligned} P(X > 6) &= \frac{8-6}{8-2} \\ P(X > 6) &= \frac{2}{6} \end{aligned}$$

$$P(X > 6) = 0.3333$$

Therefore:

$$33.33\%$$

Final Answers

Measure	Result
Mean	5 minutes
Variance	3 min ²
Standard deviation	1.732 minutes

Measure	Result
$P(3 \leq X \leq 5)$	33.33%
$P(X > 6)$	33.33%

Data Science Interpretation

A Data Scientist can use a Uniform distribution to **simulate synthetic waiting-time data** when only a lower and upper bound are known and there is a reasonable assumption that values inside that range are equally likely.

★ Important Uniform Distribution Formulas

For:

$$X \sim U(a, b)$$

PDF

$$f(x) = \frac{1}{b-a}, \quad a \leq x \leq b$$

Probability between two values

$$P(c \leq X \leq d) = \frac{d-c}{b-a}$$

where:

$$a \leq c < d \leq b$$

Cumulative Distribution Function

$$F(x) = \frac{x-a}{b-a}$$

for:

$$a \leq x \leq b$$

Mean

$$E(X) = \frac{a + b}{2}$$

Variance

$$Var(X) = \frac{(b - a)^2}{12}$$

Standard Deviation

$$\sigma = \frac{b - a}{\sqrt{12}}$$



Uniform Distribution in Data Science

Application	Example
Random Sampling	Generate random values between 0 and 1
Simulation	Simulate uncertain waiting times
Monte Carlo Methods	Generate random inputs
ML Initialization	Generate initial parameters within a bounded interval
A/B Testing	Randomly assign users
Synthetic Data	Generate bounded numerical features
Data Augmentation	Introduce controlled random variations
Algorithm Testing	Generate random test inputs
Feature Simulation	Create values within known minimum and maximum



Key Intuition

The simplest way to remember Uniform Distribution is:

"Every value within the specified interval is equally likely in terms of probability density."

For example:

$$X \sim U(10, 20)$$

The interval has length:

$$20 - 10 = 10$$

So the probability of falling between 12 and 15 is simply the **length of the desired interval divided by the total interval length**:

$$P(12 \leq X \leq 15) = \frac{15 - 12}{20 - 10}$$
$$P(12 \leq X \leq 15) = \frac{3}{10}$$

30%

The four distributions so far

Bernoulli → ONE trial

Binomial → Number of successes in fixed trials

Geometric → Trials until first success

Poisson → Number of events in a fixed interval